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PostgreSQL 16 Concurrency Control

Chapter 13 of the official PostgreSQL documentation

Sections 13.1-13.7, including SQL examples and Tables 13.1-13.3.

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https://www.postgresql.org/docs/16/mvcc.html

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Chapter 13. Concurrency Control

This chapter describes the behavior of the PostgreSQL database system when two or more sessions try to access the same data at the same time. The goals in that situation are to allow efficient access for all sessions while maintaining strict data integrity. Every developer of database applications should be familiar with the topics covered in this chapter.

https://www.postgresql.org/docs/16/mvcc-intro.html

13.1. Introduction

PostgreSQL provides a rich set of tools for developers to manage concurrent access to data. Internally, data consistency is maintained by using a multiversion model (Multiversion Concurrency Control, MVCC). This means that each SQL statement sees a snapshot of data (a database version) as it was some time ago, regardless of the current state of the underlying data. This prevents statements from viewing inconsistent data produced by concurrent transactions performing updates on the same data rows, providing transaction isolation for each database session. MVCC, by eschewing the locking methodologies of traditional database systems, minimizes lock contention in order to allow for reasonable performance in multiuser environments.

The main advantage of using the MVCC model of concurrency control rather than locking is that in MVCC locks acquired for querying (reading) data do not conflict with locks acquired for writing data, and so reading never blocks writing and writing never blocks reading. PostgreSQL maintains this guarantee even when providing the strictest level of transaction isolation through the use of an innovative Serializable Snapshot Isolation (SSI) level.

Table- and row-level locking facilities are also available in PostgreSQL for applications which don't generally need full transaction isolation and prefer to explicitly manage particular points of conflict. However, proper use of MVCC will generally provide better performance than locks. In addition, application-defined advisory locks provide a mechanism for acquiring locks that are not tied to a single transaction.

https://www.postgresql.org/docs/16/transaction-iso.html

13.2. Transaction Isolation

The SQL standard defines four levels of transaction isolation. The most strict is Serializable, which is defined by the standard in a paragraph which says that any concurrent execution of a set of Serializable transactions is guaranteed to produce the same effect as running them one at a time in some order. The other three levels are defined in terms of phenomena, resulting from interaction between concurrent transactions, which must not occur at each level. The standard notes that due to the definition of Serializable, none of these phenomena are possible at that level. (This is hardly surprising -- if the effect of the transactions must be consistent with having been run one at a time, how could you see any phenomena caused by interactions?)

The phenomena which are prohibited at various levels are:

dirty read

A transaction reads data written by a concurrent uncommitted transaction.

nonrepeatable read

A transaction re-reads data it has previously read and finds that data has been modified by another transaction (that committed since the initial read).

phantom read

A transaction re-executes a query returning a set of rows that satisfy a search condition and finds that the set of rows satisfying the condition has changed due to another recently-committed transaction.

serialization anomaly

The result of successfully committing a group of transactions is inconsistent with all possible orderings of running those transactions one at a time.

The SQL standard and PostgreSQL-implemented transaction isolation levels are described in Table 13.1.

Table 13.1. Transaction Isolation Levels

Isolation Level	Dirty Read	Nonrepeatable Read	Phantom Read	Serialization Anomaly
Read uncommitted	Allowed, but not in PG	Possible	Possible	Possible
Read committed	Not possible	Possible	Possible	Possible
Repeatable read	Not possible	Not possible	Allowed, but not in PG	Possible
Serializable	Not possible	Not possible	Not possible	Not possible

In PostgreSQL, you can request any of the four standard transaction isolation levels, but internally only three distinct isolation levels are implemented, i.e., PostgreSQL's Read Uncommitted mode behaves like Read Committed. This is because it is the only sensible way to map the standard isolation levels to PostgreSQL's multiversion concurrency control architecture.

The table also shows that PostgreSQL's Repeatable Read implementation does not allow phantom reads. This is acceptable under the SQL standard because the standard specifies which anomalies must not occur at certain isolation levels; higher guarantees are acceptable. The behavior of the available isolation levels is detailed in the following subsections.

To set the transaction isolation level of a transaction, use the command SET TRANSACTION.

Important

Some PostgreSQL data types and functions have special rules regarding transactional behavior. In particular, changes made to a sequence (and therefore the counter of a column declared using serial) are immediately visible to all other transactions and are not rolled back if the transaction that made the changes aborts. See Section 9.17 and Section 8.1.4.

13.2.1. Read Committed Isolation Level

Read Committed is the default isolation level in PostgreSQL. When a transaction uses this isolation level, a SELECT query (without a FOR UPDATE/SHARE clause) sees only data committed before the query began; it never sees either uncommitted data or changes committed by concurrent transactions during the query's execution. In effect, a SELECT query sees a snapshot of the database as of the instant the query begins to run. However, SELECT does see the effects of previous updates executed within its own transaction, even though they are not yet committed. Also note that two successive SELECT commands can see different data, even though they are within a single transaction, if other transactions commit changes after the first SELECT starts and before the second SELECT starts.

UPDATE, DELETE, SELECT FOR UPDATE, and SELECT FOR SHARE commands behave the same as SELECT in terms of searching for target rows: they will only find target rows that were committed as of the command start time. However, such a target row might have already been updated (or deleted or locked) by another concurrent transaction by the time it is found. In this case, the would-be updater will wait for the first updating transaction to commit or roll back (if it is still in progress). If the first updater rolls back, then its effects are negated and the second updater can proceed with updating the originally found row. If the first updater commits, the second updater will ignore the row if the first updater deleted it, otherwise it will attempt to apply its operation to the updated version of the row. The search condition of the command (the WHERE clause) is re-evaluated to see if the updated version of the row still matches the search condition. If so, the second updater proceeds with its operation using the updated version of the row. In the case of SELECT FOR UPDATE and SELECT FOR SHARE, this means it is the updated version of the row that is locked and returned to the client.

INSERT with an ON CONFLICT DO UPDATE clause behaves similarly. In Read Committed mode, each row proposed for insertion will either insert or update. Unless there are unrelated errors, one of those two outcomes is guaranteed. If a conflict originates in another transaction whose effects are not yet visible to the INSERT, the UPDATE clause will affect that row, even though possibly no version of that row is conventionally visible to the command.

INSERT with an ON CONFLICT DO NOTHING clause may have insertion not proceed for a row due to the outcome of another transaction whose effects are not visible to the INSERT snapshot. Again, this is only the case in Read Committed mode.

MERGE allows the user to specify various combinations of INSERT, UPDATE and DELETE subcommands. A MERGE command with both INSERT and UPDATE subcommands looks similar to INSERT with an ON CONFLICT DO UPDATE clause but does not guarantee that either INSERT or UPDATE will occur. If MERGE attempts an UPDATE or DELETE and the row is concurrently updated but the join condition still passes for the current target and the current source tuple, then MERGE will behave the same as the UPDATE or DELETE commands and perform its action on the updated version of the row. However, because MERGE can specify several actions and they can be conditional, the conditions for each action are re-evaluated on the updated

version of the row, starting from the first action, even if the action that had originally matched appears later in the list of actions. On the other hand, if the row is concurrently updated or deleted so that the join condition fails, then MERGE will evaluate the condition's NOT MATCHED actions next, and execute the first one that succeeds. If MERGE attempts an INSERT and a unique index is present and a duplicate row is concurrently inserted, then a uniqueness violation error is raised; MERGE does not attempt to avoid such errors by restarting evaluation of MATCHED conditions.

Because of the above rules, it is possible for an updating command to see an inconsistent snapshot: it can see the effects of concurrent updating commands on the same rows it is trying to update, but it does not see effects of those commands on other rows in the database. This behavior makes Read Committed mode unsuitable for commands that involve complex search conditions; however, it is just right for simpler cases. For example, consider updating bank balances with transactions like:

```
BEGIN;
UPDATE accounts SET balance = balance + 100.00 WHERE acctnum = 12345;
UPDATE accounts SET balance = balance - 100.00 WHERE acctnum = 7534;
COMMIT;
```

If two such transactions concurrently try to change the balance of account 12345, we clearly want the second transaction to start with the updated version of the account's row. Because each command is affecting only a predetermined row, letting it see the updated version of the row does not create any troublesome inconsistency.

More complex usage can produce undesirable results in Read Committed mode. For example, consider a DELETE command operating on data that is being both added and removed from its restriction criteria by another command, e.g., assume website is a two-row table with website.hits equaling 9 and 10:

```
BEGIN;
UPDATE website SET hits = hits + 1;
-- run from another session:  DELETE FROM website WHERE hits = 10;
COMMIT;
```

The DELETE will have no effect even though there is a website.hits = 10 row before and after the UPDATE. This occurs because the pre-update row value 9 is skipped, and when the UPDATE completes and DELETE obtains a lock, the new row value is no longer 10 but 11, which no longer matches the criteria.

Because Read Committed mode starts each command with a new snapshot that includes all transactions committed up to that instant, subsequent commands in the same transaction will see the effects of the committed concurrent transaction in any case. The point at issue above is whether or not a single command sees an absolutely consistent view of the database.

The partial transaction isolation provided by Read Committed mode is adequate for many applications, and this mode is fast and simple to use; however, it is not sufficient for all cases. Applications that do complex queries and updates might require a more rigorously consistent view of the database than Read Committed mode provides.

13.2.2. Repeatable Read Isolation Level

The Repeatable Read isolation level only sees data committed before the transaction began; it never sees either uncommitted data or changes committed by concurrent transactions during the transaction's execution. (However, each query does see the effects of previous updates executed within its own transaction, even though they are not yet committed.) This is a stronger guarantee than is required by the SQL standard for this isolation level, and prevents all of the phenomena described in Table 13.1 except for serialization anomalies. As mentioned above, this is specifically allowed by the standard, which only describes the minimum protections each isolation level must provide.

This level is different from Read Committed in that a query in a repeatable read transaction sees a snapshot as of the start of the first non-transaction-control statement in the transaction, not as of the start of the current statement within the transaction. Thus, successive SELECT commands within a single transaction see the same data, i.e., they do not see changes made by other transactions that committed after their own transaction started.

Applications using this level must be prepared to retry transactions due to serialization failures.

UPDATE, DELETE, MERGE, SELECT FOR UPDATE, and SELECT FOR SHARE commands behave the same as SELECT in terms of searching for target rows: they will only find target rows that were committed as of the transaction start time. However, such a target row might have already been updated (or deleted or locked) by another concurrent transaction by the time it is found. In this case, the repeatable read transaction will wait for the first updating transaction to commit or roll back (if it is still in progress). If the first updater rolls back, then its effects are negated and the repeatable read transaction can proceed with updating the originally found row. But if the first updater commits (and actually updated or deleted the row, not just locked it) then the repeatable read transaction will be rolled back with the message

```
ERROR:  could not serialize access due to concurrent update
```

because a repeatable read transaction cannot modify or lock rows changed by other transactions after the repeatable read transaction began.

When an application receives this error message, it should abort the current transaction and retry the whole transaction from the beginning. The second time through, the transaction will see the previously-committed change as part of its initial view of the database, so there is no logical conflict in using the new version of the row as the starting point for the new transaction's update.

Note that only updating transactions might need to be retried; read-only transactions will never have serialization conflicts.

The Repeatable Read mode provides a rigorous guarantee that each transaction sees a completely stable view of the database. However, this view will not necessarily always be consistent with some serial (one at a time) execution of concurrent transactions of the same level. For example, even a read-only transaction at this level may see a control record updated to show that a batch has been completed but not see one of the detail records which is logically part of the batch because it read an earlier revision of the control record. Attempts to enforce business rules by transactions running at this isolation level are not likely to work correctly without careful use of explicit locks to block conflicting transactions.

The Repeatable Read isolation level is implemented using a technique known in academic database literature and in some other database products as Snapshot Isolation. Differences in behavior and performance may be observed when compared with systems that use a traditional locking technique that reduces concurrency. Some other systems may even offer Repeatable Read and Snapshot Isolation as distinct isolation levels with

different behavior. The permitted phenomena that distinguish the two techniques were not formalized by database researchers until after the SQL standard was developed, and are outside the scope of this manual. For a full treatment, please see [berenson95].

Note

Prior to PostgreSQL version 9.1, a request for the Serializable transaction isolation level provided exactly the same behavior described here. To retain the legacy Serializable behavior, Repeatable Read should now be requested.

13.2.3. Serializable Isolation Level

The Serializable isolation level provides the strictest transaction isolation. This level emulates serial transaction execution for all committed transactions; as if transactions had been executed one after another, serially, rather than concurrently. However, like the Repeatable Read level, applications using this level must be prepared to retry transactions due to serialization failures. In fact, this isolation level works exactly the same as Repeatable Read except that it also monitors for conditions which could make execution of a concurrent set of serializable transactions behave in a manner inconsistent with all possible serial (one at a time) executions of those transactions. This monitoring does not introduce any blocking beyond that present in repeatable read, but there is some overhead to the monitoring, and detection of the conditions which could cause a serialization anomaly will trigger a serialization failure.

As an example, consider a table mytab, initially containing:

```
 class | value
-------+-------
     1 |    10
     1 |    20
     2 |   100
     2 |   200
```

Suppose that serializable transaction A computes:

```
SELECT SUM(value) FROM mytab WHERE class = 1;
```

and then inserts the result (30) as the value in a new row with class = 2. Concurrently, serializable transaction B computes:

```
SELECT SUM(value) FROM mytab WHERE class = 2;
```

and obtains the result 300, which it inserts in a new row with class = 1. Then both transactions try to commit. If either transaction were running at the Repeatable Read isolation level, both would be allowed to commit; but since there is no serial order of execution consistent with the result, using Serializable transactions will allow one transaction to commit and will roll the other back with this message:

```
ERROR:  could not serialize access due to read/write dependencies among transactions
```

This is because if A had executed before B, B would have computed the sum 330, not 300, and similarly the other order would have resulted in a different sum computed by A.

When relying on Serializable transactions to prevent anomalies, it is important that any data read from a permanent user table not be considered valid until the transaction which read it has successfully committed. This is true even for read-only transactions, except that data read within a deferrable read-only transaction is known to be valid as soon as it is read, because such a transaction waits until it can acquire a snapshot

guaranteed to be free from such problems before starting to read any data. In all other cases applications must not depend on results read during a transaction that later aborted; instead, they should retry the transaction until it succeeds.

To guarantee true serializability PostgreSQL uses predicate locking, which means that it keeps locks which allow it to determine when a write would have had an impact on the result of a previous read from a concurrent transaction, had it run first. In PostgreSQL these locks do not cause any blocking and therefore can not play any part in causing a deadlock. They are used to identify and flag dependencies among concurrent Serializable transactions which in certain combinations can lead to serialization anomalies. In contrast, a Read Committed or Repeatable Read transaction which wants to ensure data consistency may need to take out a lock on an entire table, which could block other users attempting to use that table, or it may use SELECT FOR UPDATE or SELECT FOR SHARE which not only can block other transactions but cause disk access.

Predicate locks in PostgreSQL, like in most other database systems, are based on data actually accessed by a transaction. These will show up in the pg_locks system view with a mode of SIReadLock. The particular locks acquired during execution of a query will depend on the plan used by the query, and multiple finer-grained locks (e.g., tuple locks) may be combined into fewer coarser-grained locks (e.g., page locks) during the course of the transaction to prevent exhaustion of the memory used to track the locks. A READ ONLY transaction may be able to release its SIRead locks before completion, if it detects that no conflicts can still occur which could lead to a serialization anomaly. In fact, READ ONLY transactions will often be able to establish that fact at startup and avoid taking any predicate locks. If you explicitly request a SERIALIZABLE READ ONLY DEFERRABLE transaction, it will block until it can establish this fact. (This is the only case where Serializable transactions block but Repeatable Read transactions don't.) On the other hand, SIRead locks often need to be kept past transaction commit, until overlapping read write transactions complete.

Consistent use of Serializable transactions can simplify development. The guarantee that any set of successfully committed concurrent Serializable transactions will have the same effect as if they were run one at a time means that if you can demonstrate that a single transaction, as written, will do the right thing when run by itself, you can have confidence that it will do the right thing in any mix of Serializable transactions, even without any information about what those other transactions might do, or it will not successfully commit. It is important that an environment which uses this technique have a generalized way of handling serialization failures (which always return with an SQLSTATE value of '40001'), because it will be very hard to predict exactly which transactions might contribute to the read/write dependencies and need to be rolled back to prevent serialization anomalies. The monitoring of read/write dependencies has a cost, as does the restart of transactions which are terminated with a serialization failure, but balanced against the cost and blocking involved in use of explicit locks and SELECT FOR UPDATE or SELECT FOR SHARE, Serializable transactions are the best performance choice for some environments.

While PostgreSQL's Serializable transaction isolation level only allows concurrent transactions to commit if it can prove there is a serial order of execution that would produce the same effect, it doesn't always prevent errors from being raised that would not occur in true serial execution. In particular, it is possible to see unique constraint violations caused by conflicts with overlapping Serializable transactions even after explicitly checking that the key isn't present before attempting to insert it. This can be avoided by making sure that all Serializable transactions that insert potentially conflicting keys explicitly check if they can do so first. For example, imagine an application that asks the user for a new key and then checks that it doesn't exist already by trying to select it first, or generates a new key by selecting the maximum existing key and adding one. If some Serializable transactions insert new keys directly without following this protocol, unique constraints violations might be reported even in cases where they could not occur in a serial execution of the concurrent

transactions.

For optimal performance when relying on Serializable transactions for concurrency control, these issues should be considered:

Declare transactions as READ ONLY when possible.

Control the number of active connections, using a connection pool if needed. This is always an important performance consideration, but it can be particularly important in a busy system using Serializable transactions.

Don't put more into a single transaction than needed for integrity purposes.

Don't leave connections dangling "idle in transaction" longer than necessary. The configuration parameter idle_in_transaction_session_timeout may be used to automatically disconnect lingering sessions.

Eliminate explicit locks, SELECT FOR UPDATE, and SELECT FOR SHARE where no longer needed due to the protections automatically provided by Serializable transactions.

When the system is forced to combine multiple page-level predicate locks into a single relation-level predicate lock because the predicate lock table is short of memory, an increase in the rate of serialization failures may occur. You can avoid this by increasing max_pred_locks_per_transaction, max_pred_locks_per_relation, and/or max_pred_locks_per_page.

A sequential scan will always necessitate a relation-level predicate lock. This can result in an increased rate of serialization failures. It may be helpful to encourage the use of index scans by reducing random_page_cost and/or increasing cpu_tuple_cost. Be sure to weigh any decrease in transaction rollbacks and restarts against any overall change in query execution time.

The Serializable isolation level is implemented using a technique known in academic database literature as Serializable Snapshot Isolation, which builds on Snapshot Isolation by adding checks for serialization anomalies. Some differences in behavior and performance may be observed when compared with other systems that use a traditional locking technique. Please see [ports12] for detailed information.

https://www.postgresql.org/docs/16/explicit-locking.html

13.3. Explicit Locking

PostgreSQL provides various lock modes to control concurrent access to data in tables. These modes can be used for application-controlled locking in situations where MVCC does not give the desired behavior. Also, most PostgreSQL commands automatically acquire locks of appropriate modes to ensure that referenced tables are not dropped or modified in incompatible ways while the command executes. (For example, TRUNCATE cannot safely be executed concurrently with other operations on the same table, so it obtains an ACCESS EXCLUSIVE lock on the table to enforce that.)

To examine a list of the currently outstanding locks in a database server, use the pg_locks system view. For more information on monitoring the status of the lock manager subsystem, refer to Chapter 28.

13.3.1. Table-Level Locks

The list below shows the available lock modes and the contexts in which they are used automatically by PostgreSQL. You can also acquire any of these locks explicitly with the command LOCK. Remember that all of these lock modes are table-level locks, even if the name contains the word "row"; the names of the lock modes are historical. To some extent the names reflect the typical usage of each lock mode - but the semantics are all the same. The only real difference between one lock mode and another is the set of lock modes with which each conflicts (see Table 13.2). Two transactions cannot hold locks of conflicting modes on the same table at the same time. (However, a transaction never conflicts with itself. For example, it might acquire ACCESS EXCLUSIVE lock and later acquire ACCESS SHARE lock on the same table.)
Non-conflicting lock modes can be held concurrently by many transactions. Notice in particular that some lock modes are self-conflicting (for example, an ACCESS EXCLUSIVE lock cannot be held by more than one transaction at a time) while others are not self-conflicting (for example, an ACCESS SHARE lock can be held by multiple transactions).

Table-Level Lock Modes

ACCESS SHARE (AccessShareLock)

Conflicts with the ACCESS EXCLUSIVE lock mode only.

The SELECT command acquires a lock of this mode on referenced tables. In general, any query that only reads a table and does not modify it will acquire this lock mode.

ROW SHARE (RowShareLock)

Conflicts with the EXCLUSIVE and ACCESS EXCLUSIVE lock modes.

The SELECT command acquires a lock of this mode on all tables on which one of the FOR UPDATE, FOR NO KEY UPDATE, FOR SHARE, or FOR KEY SHARE options is specified (in addition to ACCESS SHARE locks on any other tables that are referenced without any explicit FOR ... locking option).

ROW EXCLUSIVE (RowExclusiveLock)

Conflicts with the SHARE, SHARE ROW EXCLUSIVE, EXCLUSIVE, and ACCESS EXCLUSIVE lock modes.

The commands UPDATE, DELETE, INSERT, and MERGE acquire this lock mode on the target table (in addition to ACCESS SHARE locks on any other referenced tables). In general, this lock mode will be acquired by any command that modifies data in a table.

SHARE UPDATE EXCLUSIVE (ShareUpdateExclusiveLock)

Conflicts with the SHARE UPDATE EXCLUSIVE, SHARE, SHARE ROW EXCLUSIVE, EXCLUSIVE, and ACCESS EXCLUSIVE lock modes. This mode protects a table against concurrent schema changes and VACUUM runs.

Acquired by VACUUM (without FULL), ANALYZE, CREATE INDEX CONCURRENTLY, CREATE STATISTICS, COMMENT ON, REINDEX CONCURRENTLY, and certain ALTER INDEX and ALTER TABLE variants (for full details see the documentation of these commands).

SHARE (ShareLock)

Conflicts with the ROW EXCLUSIVE, SHARE UPDATE EXCLUSIVE, SHARE ROW EXCLUSIVE, EXCLUSIVE, and ACCESS EXCLUSIVE lock modes. This mode protects a table against concurrent data changes.

Acquired by CREATE INDEX (without CONCURRENTLY).

SHARE ROW EXCLUSIVE (ShareRowExclusiveLock)

Conflicts with the ROW EXCLUSIVE, SHARE UPDATE EXCLUSIVE, SHARE, SHARE ROW EXCLUSIVE, EXCLUSIVE, and ACCESS EXCLUSIVE lock modes. This mode protects a table against concurrent data changes, and is self-exclusive so that only one session can hold it at a time.

Acquired by CREATE TRIGGER and some forms of ALTER TABLE.

EXCLUSIVE (ExclusiveLock)

Conflicts with the ROW SHARE, ROW EXCLUSIVE, SHARE UPDATE EXCLUSIVE, SHARE, SHARE ROW EXCLUSIVE, EXCLUSIVE, and ACCESS EXCLUSIVE lock modes. This mode allows only concurrent ACCESS SHARE locks, i.e., only reads from the table can proceed in parallel with a transaction holding this lock mode.

Acquired by REFRESH MATERIALIZED VIEW CONCURRENTLY.

ACCESS EXCLUSIVE (AccessExclusiveLock)

Conflicts with locks of all modes (ACCESS SHARE, ROW SHARE, ROW EXCLUSIVE, SHARE UPDATE EXCLUSIVE, SHARE, SHARE ROW EXCLUSIVE, EXCLUSIVE, and ACCESS EXCLUSIVE). This mode guarantees that the holder is the only transaction accessing the table in any way.

Acquired by the DROP TABLE, TRUNCATE, REINDEX, CLUSTER, VACUUM FULL, and REFRESH MATERIALIZED VIEW (without CONCURRENTLY) commands. Many forms of ALTER INDEX and ALTER TABLE also acquire a lock at this level. This is also the default lock mode for LOCK TABLE statements that do not specify a mode explicitly.

Tip

Only an ACCESS EXCLUSIVE lock blocks a SELECT (without FOR UPDATE/SHARE) statement.

Once acquired, a lock is normally held until the end of the transaction. But if a lock is acquired after establishing a savepoint, the lock is released immediately if the savepoint is rolled back to. This is consistent with the principle that ROLLBACK cancels all effects of the commands since the savepoint. The same holds for locks acquired within a PL/pgSQL exception block: an error escape from the block releases locks acquired within it.

Table 13.2. Conflicting Lock Modes

Requested Lock Mode	ACCESS SHARE	ROW SHARE	ROW EXCL.	SHARE UPDATE EXCL.	SHARE	SHARE ROW EXCL.	EXCL.	ACCESS EXCL.
ACCESS SHARE								X
ROW SHARE							X	X
ROW EXCL.					X	X	X	X
SHARE UPDATE EXCL.				X	X	X	X	X
SHARE			X	X		X	X	X
SHARE ROW EXCL.			X	X	X	X	X	X
EXCL.		X	X	X	X	X	X	X
ACCESS EXCL.	X	X	X	X	X	X	X	X

13.3.2. Row-Level Locks

In addition to table-level locks, there are row-level locks, which are listed as below with the contexts in which they are used automatically by PostgreSQL. See Table 13.3 for a complete table of row-level lock conflicts. Note that a transaction can hold conflicting locks on the same row, even in different subtransactions; but other than that, two transactions can never hold conflicting locks on the same row. Row-level locks do not affect data querying; they block only writers and lockers to the same row. Row-level locks are released at transaction end or during savepoint rollback, just like table-level locks.

Row-Level Lock Modes

FOR UPDATE

FOR UPDATE causes the rows retrieved by the SELECT statement to be locked as though for update. This prevents them from being locked, modified or deleted by other transactions until the current transaction ends. That is, other transactions that attempt UPDATE, DELETE, SELECT FOR UPDATE, SELECT FOR NO KEY UPDATE, SELECT FOR SHARE or SELECT FOR KEY SHARE of these rows will be blocked until the current transaction ends; conversely, SELECT FOR UPDATE will wait for a concurrent transaction that has run any of those commands on the same row, and will then lock and return the updated row (or no row, if the row was deleted). Within a REPEATABLE READ or SERIALIZABLE transaction, however, an error will be thrown if a row to be locked has changed since the transaction started. For further discussion see Section 13.4.

The FOR UPDATE lock mode is also acquired by any DELETE on a row, and also by an UPDATE that modifies the values of certain columns. Currently, the set of columns considered for the UPDATE case are those that have a unique index on them that can be used in a foreign key (so partial indexes and expressional indexes are not considered), but this may change in the future.

FOR NO KEY UPDATE

Behaves similarly to FOR UPDATE, except that the lock acquired is weaker: this lock will not block SELECT FOR KEY SHARE commands that attempt to acquire a lock on the same rows. This lock mode is also acquired by any UPDATE that does not acquire a FOR UPDATE lock.

FOR SHARE

Behaves similarly to FOR NO KEY UPDATE, except that it acquires a shared lock rather than exclusive lock on each retrieved row. A shared lock blocks other transactions from performing UPDATE, DELETE, SELECT FOR UPDATE or SELECT FOR NO KEY UPDATE on these rows, but it does not prevent them from performing SELECT FOR SHARE or SELECT FOR KEY SHARE.

FOR KEY SHARE

Behaves similarly to FOR SHARE, except that the lock is weaker: SELECT FOR UPDATE is blocked, but not SELECT FOR NO KEY UPDATE. A key-shared lock blocks other transactions from performing DELETE or any UPDATE that changes the key values, but not other UPDATE, and neither does it prevent SELECT FOR NO KEY UPDATE, SELECT FOR SHARE, or SELECT FOR KEY SHARE.

PostgreSQL doesn't remember any information about modified rows in memory, so there is no limit on the number of rows locked at one time. However, locking a row might cause a disk write, e.g., SELECT FOR UPDATE modifies selected rows to mark them locked, and so will result in disk writes.

Table 13.3. Conflicting Row-Level Locks

Requested Lock Mode	FOR KEY SHARE	FOR SHARE	FOR NO KEY UPDATE	FOR UPDATE
FOR KEY SHARE				X
FOR SHARE			X	X
FOR NO KEY UPDATE		X	X	X
FOR UPDATE	X	X	X	X

13.3.3. Page-Level Locks

In addition to table and row locks, page-level share/exclusive locks are used to control read/write access to table pages in the shared buffer pool. These locks are released immediately after a row is fetched or updated. Application developers normally need not be concerned with page-level locks, but they are mentioned here for completeness.

13.3.4. Deadlocks

The use of explicit locking can increase the likelihood of deadlocks, wherein two (or more) transactions each hold locks that the other wants. For example, if transaction 1 acquires an exclusive lock on table A and then tries to acquire an exclusive lock on table B, while transaction 2 has already exclusive-locked table B and now wants an exclusive lock on table A, then neither one can proceed. PostgreSQL automatically detects deadlock situations and resolves them by aborting one of the transactions involved, allowing the other(s) to complete. (Exactly which transaction will be aborted is difficult to predict and should not be relied upon.)

Note that deadlocks can also occur as the result of row-level locks (and thus, they can occur even if explicit locking is not used). Consider the case in which two concurrent transactions modify a table. The first transaction executes:

```
UPDATE accounts SET balance = balance + 100.00 WHERE acctnum = 11111;
```

This acquires a row-level lock on the row with the specified account number. Then, the second transaction executes:

```
UPDATE accounts SET balance = balance + 100.00 WHERE acctnum = 22222;
UPDATE accounts SET balance = balance - 100.00 WHERE acctnum = 11111;
```

The first UPDATE statement successfully acquires a row-level lock on the specified row, so it succeeds in updating that row. However, the second UPDATE statement finds that the row it is attempting to update has already been locked, so it waits for the transaction that acquired the lock to complete. Transaction two is now waiting on transaction one to complete before it continues execution. Now, transaction one executes:

```
UPDATE accounts SET balance = balance - 100.00 WHERE acctnum = 22222;
```

Transaction one attempts to acquire a row-level lock on the specified row, but it cannot: transaction two already holds such a lock. So it waits for transaction two to complete. Thus, transaction one is blocked on transaction two, and transaction two is blocked on transaction one: a deadlock condition. PostgreSQL will detect this situation and abort one of the transactions.

The best defense against deadlocks is generally to avoid them by being certain that all applications using a database acquire locks on multiple objects in a consistent order. In the example above, if both transactions had updated the rows in the same order, no deadlock would have occurred. One should also ensure that the first lock acquired on an object in a transaction is the most restrictive mode that will be needed for that object. If it is not feasible to verify this in advance, then deadlocks can be handled on-the-fly by retrying transactions that abort due to deadlocks.

So long as no deadlock situation is detected, a transaction seeking either a table-level or row-level lock will wait indefinitely for conflicting locks to be released. This means it is a bad idea for applications to hold transactions open for long periods of time (e.g., while waiting for user input).

13.3.5. Advisory Locks

PostgreSQL provides a means for creating locks that have application-defined meanings. These are called advisory locks, because the system does not enforce their use - it is up to the application to use them correctly. Advisory locks can be useful for locking strategies that are an awkward fit for the MVCC model. For example, a common use of advisory locks is to emulate pessimistic locking strategies typical of so-called "flat file" data management systems. While a flag stored in a table could be used for the same purpose, advisory locks are faster, avoid table bloat, and are automatically cleaned up by the server at the end of the session.

There are two ways to acquire an advisory lock in PostgreSQL: at session level or at transaction level. Once acquired at session level, an advisory lock is held until explicitly released or the session ends. Unlike standard lock requests, session-level advisory lock requests do not honor transaction semantics: a lock acquired during a transaction that is later rolled back will still be held following the rollback, and likewise an unlock is effective even if the calling transaction fails later. A lock can be acquired multiple times by its owning process; for each completed lock request there must be a corresponding unlock request before the lock is actually released. Transaction-level lock requests, on the other hand, behave more like regular lock requests: they are automatically released at the end of the transaction, and there is no explicit unlock operation. This behavior is often more convenient than the session-level behavior for short-term usage of an advisory lock. Session-level and transaction-level lock requests for the same advisory lock identifier will block each other in the expected way. If a session already holds a given advisory lock, additional requests by it will always succeed, even if other sessions are awaiting the lock; this statement is true regardless of whether the existing lock hold and new request are at session level or transaction level.

Like all locks in PostgreSQL, a complete list of advisory locks currently held by any session can be found in the pg_locks system view.

Both advisory locks and regular locks are stored in a shared memory pool whose size is defined by the configuration variables max_locks_per_transaction and max_connections. Care must be taken not to exhaust this memory or the server will be unable to grant any locks at all. This imposes an upper limit on the number of advisory locks grantable by the server, typically in the tens to hundreds of thousands depending on how the server is configured.

In certain cases using advisory locking methods, especially in queries involving explicit ordering and LIMIT clauses, care must be taken to control the locks acquired because of the order in which SQL expressions are evaluated. For example:

```
SELECT pg_advisory_lock(id) FROM foo WHERE id = 12345; -- ok
SELECT pg_advisory_lock(id) FROM foo WHERE id > 12345 LIMIT 100; -- danger!
SELECT pg_advisory_lock(q.id) FROM
(
  SELECT id FROM foo WHERE id > 12345 LIMIT 100
) q; -- ok
```

In the above queries, the second form is dangerous because the LIMIT is not guaranteed to be applied before the locking function is executed. This might cause some locks to be acquired that the application was not expecting, and hence would fail to release (until it ends the session). From the point of view of the application, such locks would be dangling, although still viewable in pg_locks.

The functions provided to manipulate advisory locks are described in Section 9.27.10.

https://www.postgresql.org/docs/16/applevel-consistency.html

13.4. Data Consistency Checks at the Application Level

It is very difficult to enforce business rules regarding data integrity using Read Committed transactions because the view of the data is shifting with each statement, and even a single statement may not restrict itself to the statement's snapshot if a write conflict occurs.

While a Repeatable Read transaction has a stable view of the data throughout its execution, there is a subtle issue with using MVCC snapshots for data consistency checks, involving something known as read/write conflicts. If one transaction writes data and a concurrent transaction attempts to read the same data (whether before or after the write), it cannot see the work of the other transaction. The reader then appears to have executed first regardless of which started first or which committed first. If that is as far as it goes, there is no problem, but if the reader also writes data which is read by a concurrent transaction there is now a transaction which appears to have run before either of the previously mentioned transactions. If the transaction which appears to have executed last actually commits first, it is very easy for a cycle to appear in a graph of the order of execution of the transactions. When such a cycle appears, integrity checks will not work correctly without some help.

As mentioned in Section 13.2.3, Serializable transactions are just Repeatable Read transactions which add nonblocking monitoring for dangerous patterns of read/write conflicts. When a pattern is detected which could cause a cycle in the apparent order of execution, one of the transactions involved is rolled back to break the cycle.

13.4.1. Enforcing Consistency with Serializable Transactions

If the Serializable transaction isolation level is used for all writes and for all reads which need a consistent view of the data, no other effort is required to ensure consistency. Software from other environments which is written to use serializable transactions to ensure consistency should "just work" in this regard in PostgreSQL.

When using this technique, it will avoid creating an unnecessary burden for application programmers if the application software goes through a framework which automatically retries transactions which are rolled back with a serialization failure. It may be a good idea to set default_transaction_isolation to serializable. It would also be wise to take some action to ensure that no other transaction isolation level is used, either inadvertently or to subvert integrity checks, through checks of the transaction isolation level in triggers.

See Section 13.2.3 for performance suggestions.

Warning: Serializable Transactions and Data Replication

This level of integrity protection using Serializable transactions does not yet extend to hot standby mode (Section 27.4) or logical replicas. Because of that, those using hot standby or logical replication may want to use Repeatable Read and explicit locking on the primary.

13.4.2. Enforcing Consistency with Explicit Blocking Locks

When non-serializable writes are possible, to ensure the current validity of a row and protect it against concurrent updates one must use SELECT FOR UPDATE, SELECT FOR SHARE, or an appropriate LOCK TABLE statement. (SELECT FOR UPDATE and SELECT FOR SHARE lock just the returned rows against concurrent updates, while LOCK TABLE locks the whole table.) This should be taken into account when porting applications to PostgreSQL from other environments.

Also of note to those converting from other environments is the fact that SELECT FOR UPDATE does not ensure that a concurrent transaction will not update or delete a selected row. To do that in PostgreSQL you must actually update the row, even if no values need to be changed. SELECT FOR UPDATE temporarily blocks other transactions from acquiring the same lock or executing an UPDATE or DELETE which would affect the locked row, but once the transaction holding this lock commits or rolls back, a blocked transaction will proceed with the conflicting operation unless an actual UPDATE of the row was performed while the lock was held.

Global validity checks require extra thought under non-serializable MVCC. For example, a banking application might wish to check that the sum of all credits in one table equals the sum of debits in another table, when both tables are being actively updated. Comparing the results of two successive SELECT sum(...) commands will not work reliably in Read Committed mode, since the second query will likely include the results of transactions not counted by the first. Doing the two sums in a single repeatable read transaction will give an accurate picture of only the effects of transactions that committed before the repeatable read transaction started - but one might legitimately wonder whether the answer is still relevant by the time it is delivered. If the repeatable read transaction itself applied some changes before trying to make the consistency check, the usefulness of the check becomes even more debatable, since now it includes some but not all post-transaction-start changes. In such cases a careful person might wish to lock all tables needed for the check, in order to get an indisputable picture of current reality. A SHARE mode (or higher) lock guarantees that there are no uncommitted changes in the locked table, other than those of the current transaction.

Note also that if one is relying on explicit locking to prevent concurrent changes, one should either use Read Committed mode, or in Repeatable Read mode be careful to obtain locks before performing queries. A lock obtained by a repeatable read transaction guarantees that no other transactions modifying the table are still running, but if the snapshot seen by the transaction predates obtaining the lock, it might predate some now-committed changes in the table. A repeatable read transaction's snapshot is actually frozen at the start of its first query or data-modification command (SELECT, INSERT, UPDATE, DELETE, or MERGE), so it is possible to obtain locks explicitly before the snapshot is frozen.

https://www.postgresql.org/docs/16/mvcc-serialization-failure-handling.html

13.5. Serialization Failure Handling

Both Repeatable Read and Serializable isolation levels can produce errors that are designed to prevent serialization anomalies. As previously stated, applications using these levels must be prepared to retry transactions that fail due to serialization errors. Such an error's message text will vary according to the precise circumstances, but it will always have the SQLSTATE code 40001 (serialization_failure).

It may also be advisable to retry deadlock failures. These have the SQLSTATE code 40P01 (deadlock_detected).

In some cases it is also appropriate to retry unique-key failures, which have SQLSTATE code 23505 (unique_violation), and exclusion constraint failures, which have SQLSTATE code 23P01 (exclusion_violation). For example, if the application selects a new value for a primary key column after inspecting the currently stored keys, it could get a unique-key failure because another application instance selected the same new key concurrently. This is effectively a serialization failure, but the server will not detect it as such because it cannot "see" the connection between the inserted value and the previous reads. There are also some corner cases in which the server will issue a unique-key or exclusion constraint error even though in principle it has enough information to determine that a serialization problem is the underlying cause. While it's recommendable to just retry serialization_failure errors unconditionally, more care is needed when retrying these other error codes, since they might represent persistent error conditions rather than transient failures.

It is important to retry the complete transaction, including all logic that decides which SQL to issue and/or which values to use. Therefore, PostgreSQL does not offer an automatic retry facility, since it cannot do so with any guarantee of correctness.

Transaction retry does not guarantee that the retried transaction will complete; multiple retries may be needed. In cases with very high contention, it is possible that completion of a transaction may take many attempts. In cases involving a conflicting prepared transaction, it may not be possible to make progress until the prepared transaction commits or rolls back.

https://www.postgresql.org/docs/16/mvcc-caveats.html

13.6. Caveats

Some DDL commands, currently only TRUNCATE and the table-rewriting forms of ALTER TABLE, are not MVCC-safe. This means that after the truncation or rewrite commits, the table will appear empty to concurrent transactions, if they are using a snapshot taken before the DDL command committed. This will only be an issue for a transaction that did not access the table in question before the DDL command started - any transaction that has done so would hold at least an ACCESS SHARE table lock, which would block the DDL command until that transaction completes. So these commands will not cause any apparent inconsistency in the table contents for successive queries on the target table, but they could cause visible inconsistency between the contents of the target table and other tables in the database.

Support for the Serializable transaction isolation level has not yet been added to hot standby replication targets (described in Section 27.4). The strictest isolation level currently supported in hot standby mode is Repeatable Read. While performing all permanent database writes within Serializable transactions on the primary will ensure that all standbys will eventually reach a consistent state, a Repeatable Read transaction run on the standby can sometimes see a transient state that is inconsistent with any serial execution of the transactions on the primary.

Internal access to the system catalogs is not done using the isolation level of the current transaction. This means that newly created database objects such as tables are visible to concurrent Repeatable Read and Serializable transactions, even though the rows they contain are not. In contrast, queries that explicitly examine the system catalogs don't see rows representing concurrently created database objects, in the higher isolation levels.

https://www.postgresql.org/docs/16/locking-indexes.html

13.7. Locking and Indexes

Though PostgreSQL provides nonblocking read/write access to table data, nonblocking read/write access is not currently offered for every index access method implemented in PostgreSQL. The various index types are handled as follows:

B-tree, GiST and SP-GiST indexes

Short-term share/exclusive page-level locks are used for read/write access. Locks are released immediately after each index row is fetched or inserted. These index types provide the highest concurrency without deadlock conditions.

Hash indexes

Share/exclusive hash-bucket-level locks are used for read/write access. Locks are released after the whole bucket is processed. Bucket-level locks provide better concurrency than index-level ones, but deadlock is possible since the locks are held longer than one index operation.

GIN indexes

Short-term share/exclusive page-level locks are used for read/write access. Locks are released immediately after each index row is fetched or inserted. But note that insertion of a GIN-indexed value usually produces several index key insertions per row, so GIN might do substantial work for a single value's insertion.

Currently, B-tree indexes offer the best performance for concurrent applications; since they also have more features than hash indexes, they are the recommended index type for concurrent applications that need to index scalar data. When dealing with non-scalar data, B-trees are not useful, and GiST, SP-GiST or GIN indexes should be used instead.

https://www.postgresql.org/docs/16/legalnotice.html

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